MILITARY SPECIFICATION

HEAD ASSEMBLY: 11747439
(Periscope M32CE1)

This specification is approved for use by the U.S. army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a modular unit of the Tank Gunner's M32CE1 Periscope, herein referred to as Head Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent. specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acqtisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be ofuse in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDARQA, Dover, NewJersey07801 by using the self-addressed standardization Document Improvement Proposal (DDForm 1426) appealing at the end ofthis document or by letter,

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS (see 6.4)

US Army Armament Research and Development Command (ARRADCOM)

5549108	Telescope, Collimating (3X)
8247391	Test Fixture for Periscope
8565556	Pressure Tester, Two-Station
10547987	Set-Up Instructions for Calibration Standards, Periscope M32
10548766	Calibration Standard, 26° Elevation
10548767	Support Assembly Collimator
11729723	Gage, Minimum Interchangeability
11732101	Gage, Maximum Interchangeability
11732107	Gage, Maximum Interchangeability
11732108	Gage, Minimum Interchangeability
11732155	Gage, Minimum Interchangeability
11732156	Gage, Maximum Interchangeability
11738140	Final Acceptance Inspection Equipment
11747439	Head Assembly

PACKAGING DATA SHEET

11747439	Packaging of Head Assembly 11747439

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.-The head assembly shall be manufactured in accordance with Drawing 11747439 and drawings pertaining thereto.

3.2 Interchangeability.

a. Head Assembly. with modules (Elbow Assembly, Passive and Body Assembly, Daylight). The head assembly shall be capable of the insertion and removal of the passive and daylight modules without restriction (gages are specified in lieu of modules in 4.6.12.1).

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b. Head Assembly with mount. - The head assembly shall be capable of insertion into the Mount Periscope: M118E1 (gages are specified in lieu of mount in 4.6.12.2).
shall be capable of being firmly sercured and subsequently removed.

3.3 General specification. - The contractor shall be responsible for adherence to, and compliance with the following requirements of MIL-F-13926.

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Part identification and marking

3.4 Environmental.

3.4.1 Storage temperautre. - The head assebly shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all the requirements of this specification at standard ambient temperaure (73° ± 18°F) after having been exposed and thermally stabilized at ambient temperatures of -80° to +160°F.

3.4.2 Opertating temperatures. - The head assembly shall meet the applicable requirements of 3.5 while exposed and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5 Operability, elevation input coupling. - The running torque required to rotate the input coupling through maximum elevation and depression shall not exceed 6 pound-inches at standard ambient temperature (73° to 180°F). When checked at the temperature conditions specified in 3.4.2, the running torque shall not exceed 10 pound-inches.

3.6 Orientation. -The requirements of 3.7, 3.10 and 3.11 shall be met with the head assembly positioned as follows:

a. Datum Plane “Y” on a horizontal plane
b. Side “L” of the input coupling positioned 22 degrees 30 minutes with respect to a plane defined by the three mounting pads as measured in a vertical plane.

NOTE: The following reference applies throughout this specification:

Datum Plane “W” - See Drawing 11747441

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3.7 Collimation.-The requirements of 3.7.1 to 3.7.8 inclusive, shall be met with:

a. An incident line of sight which is originally perpendicular to a plane defined by the three mounting pads.

b. The input coupling positioned in accordance with 3.6.

3.7.1 Pre vibration/Pre-shock daylight viewing area. - The incident line of sight shall emerge perpendicular to Datum Plane "Y" within 0.20 mil.

3.7.2 Pre-vibration/Pre-shock passive viewing area. - The incident line of sight shall emerge perpendicular to Datum Plane "Y" within 0.20 mil.

3.7.3 Unity power area. - The reflected line of sight shall be depressed 26 degrees below Datum Plane "Y" within 15 minutes, and be perpendicular to Datum Plane "W" within 15 minutes.

3.7.4 Excursion rnage. - Rotation of the input coupling shall cause the line of sight to deflect a minimum of 22.0 degrees above to 18.0 degrees below the horizontal plane.

3.7.4.1 Coupling travel.- Rotation of the input coupling shall be limited by contact with the coupling stop pin only.

NOTE: At no time shall the input coupling be allowed to strike hard against the travel stops.

3.7.5 Plumb travel. - Starting with the line of sight in coincidence with a vertical target line at the horizontal position, the line of sight shall track the target line within 0.25 mil when elevated and depressed through the range specified in 3.7.4.

3.7.6 Elevation accuracy.- This requirement shall be met with backlash removed and the angle measured in a vertical plane. The angle formed by the line of sight and side "L" of the input coupling shall remain constant within 0.20 mil at any elevation setting from 18 degrees depression to 22 degrees elevation.

3.7.7 Backlash. - Backlash between the line of sight and the input coupling shall not exceed 0.05 mil.

3.7.8 Image tilt. - This requirement shall be met when viewing through either the daylight or passive exit aperture. The image of

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a target line which is originally perpendicular to Datum Plane “Y” hall emerge perpendicular to Datum Plane “W” within 30 minutes of arc.

3.8 Resolution.

3.8.1 Daylight andpassive viewing areas. - Resolution shall be 10 seconds of arc or less when viewing through either the daylight or passive exit apertures with a telescope having a 1.6 inch entrance pupil. Resolution consists of detectable line structure and proper line count in all four meridians at one focus setting.

3.8.2 Unity power system. Resolution shall be 45 seconds of arc or less when viewing through the unity-power exit aperture with an observation telescope of at least 1/2 inch entrance pupil.

3.9 Vibration/Shock.

3.9.1 Vibration “A”. - Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be .5G ± 10%.

b. During a frequency sweep of 5 to 9 Hz. increase acceleration linearly from .5G to 1.5G’s ± 10%.

c. During a frequency sweep of 9 to 500 Hz, maintain constant acceleration of 1.5 G’s ± 10%.

d. During a frequency sweep of 500 to 9 Hz, maintain constant acceleration of 1.5 G’s + 10%.

e. During a frequency sweep of 9 to 5 Hz, decrease acceleration linearly from 1.5 G’s to .65 G + 10%.

f. At 5 Hz, the acceleration shall be .5 G + 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of two hours per axis. The head assembly shall show no evidence of physical failure sub-sequent to vibration. Following the vibration test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

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3.9.2 Vibration "B". - Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. The vibration shall be sinusoidal as follows:

a. At a frequency of 5 Hertz (Hz), the acceleration shall be 1.0 G ± 10%.

b. During a frequency sweep of 5 to 500 Hz, maintain constant acceleration of 1.0 G ± 10%.

c. During a frequency sweep of 500 to 5 Hz, maintain constant acceleration of 1.0 G ± 10%.

d. At 5 Hz. the acceleration shall be 1.0 G ± 10%.

The frequency of applied vibration shall be swept logarithmically in 15 minute ± 1 minute sweep cycles for a total of 15 minutes ± 1 minute along each of the three mutually perpendicular axes (total vibration time 45 minutes ± 3 minutes). The head assembly shall show no evidence of physical failure subsequent to vibration. Following the vibration test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

3.9.3 Shock. - The head assembly shall be subjected to 24 cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in each direction along each of the two mutually perpendicular horizontal axes (all four sides). The shock pulse shall have a half sine wave with a time duration of 3 ± .2 milliseconds. The head assembly shall also be subjected to six cumulative shocks consisting of three each at 35 G's ± 10%, then 20 G's ± 10% applied in one direction of along the vertical axis (normal operating position of head assembly only). The shock pulse shall have a half sine wave with a time duration of 18 ± 3 milli-seconds. The head assembly shall show no evidence of physical failure subsequent to shock. Following the shock test, the head assembly shall meet the requirements of 3.2, 3.5 through 3.8 inclusive and 3.10 through 3.13 inclusive.

3.10 Post-vibration/Post-shock, daylight viewing area. - The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.1.

3.11 Post-vibration/Post-shock, passive viewing area. - The incident line of sight shall remain within 0.10 mil of the collimation reading established in 3.7.2.

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3.12 Sealing. - The head assembly shall show no evidence of leakage inexcess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5 ± 0.10 psi for a minimum of 2 hours. The interior atmosphere of the accepted head assembly shall contain nitrogen (having a dew point of at least as low as minus 25°F at atmospheric pressure) between 0.25 and 0.50 pound per square inch gage (psig)。

3.13 Cleanliness.

a. Defects.-There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations on any glass component when viewing through either end (including the unity power system).

b. Dirt. - There shall be no more than three particles of dirt or foreign material (classed as dirt) on any glass component. The size of any dirt particle shall not exceed 0.05 inch.

3.14 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.15 Workmanship. - The head assembly shall meet the workmanship requirements of MIL-F-13926 and applicable specifications as cited on Drawing 11747439. the optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or orders the supplies may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requiremets.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 Classification of inspections. - Unless otherwise specified herein or in the contract, the following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection.

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provision of 4.3.2. Unless otherwise specified, the sample for first article approval tests shall consist of three head assemblies plus three each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The responsibility (Government or Contractor) for first article testing shall be as specified in the contract.

4.3.2 Inspections to be performed.

4.3.2.1 Head asseblies. - First article assemblies shall be tested as specified herein for all the requirements of this specification, except 3.9.2.

4.3.2.2 SQAP items. - The items covered by SQAP shall be tested as specified thereon, for all characteristics delineated on the SQAP.

4.3.3 Rejection. - If any head assembly or SQAP item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly, component, or subassembly in the sample to comply with any of the stated requirements.

4.4 Quality Conformance Inspection.

4.4.1 Inspection lot formation. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product“ provisions of MIL-STD-105.

4.4.2 Examination.-The examinations listed herein for Classification of Defects shall be performed on inspection lots as defined in 4.4.1 of this specification.

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4.4.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the Supplementary Quality Assurance Provisions (SQAP) listed in the Technical Data Package (TDP). In the absence of SQAP'S the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.4.2.2 Head assemblies. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performef on a single defect (individual characteristic) basis in accordance with MIL-STD-105, Inspection Level II, and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 100 Percent inspection. - The requirements and tests in Table I shall be inspected on a 100% basis.

TABLE I - CLASSIFICATION OF DEFECTS
(100 Percent Inspection)

CLASS	CHARACTERISTIC	REQUIREMENT	TEST PROCEDUR
CRITICAL:	NONE		
MAJOR:			
101.	Interchangeability, Head Assembly with modules (Elbow Assembly, Passive and Body Assembly, Daylight)	3.2a	4.6.12.1
102.	Interchangeability, Head Assembly with mount	3.2b	4.6.12.2
103.	Resolution, daylight and passive viewing areas	3.8.1	4.6.5
104.	Resolution, unity power system	3.8.2	4.6.6
MINOR:	None		

NOTE: The tests in Table I shall be conducted at a temperature of 73° ± 18°F.

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TABLE II - CLASSIFICATION OF DEFECTS

CLASS: CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% Defective		
105. Operability, elevation input coupling	3.5	4.6.3
106. Collimation, Pre-vibration daylignt viewing area	3.7.1	4.6.4.1
107. Collimation, Pre-vibration passive viewing area	3.7.2	4.6.4.2
108. Collimation, unity power area	3.7.3	4.6.4.3
109. Excursion range	3.7.4	4.6.4.5
110. Coupling travel	3.7.4.1	4.6.4.6
111. Plumb travel	3.7.5	4.6.4.7
112. Elevation accuracy	3.7.6	4.6.4.8
113. Backlash	3.7.7	4.6.4.4
114. Image tilt	3.7.8	4.6.4.9
115. Vibration "B"	3.9.2	4.6.7.2
116. Collimation, Post-vibration daylight viewing area	3.10	4.6.8
117. Collimation, post-vibration passive viewing area	3.11	4.6.9
118. Sealing	3.12	4.6.10
119. Cleanliness	3.13	4.6.11
120. Workmanship	3.15	Visual, Tactile
MINOR: NONE		

NOTE: The tests in Table II shall be conducted at a temperature of 73° ± 18°F.

4.4.3 Acceptance and rejection. - Rejected lots shal be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Special sampling.

4.5.1 General.-One head assembly shall be randomly selected as a sample from each 100 produced or from each month’s production, whichever occurs first. The sample shall meet the requirements and tests in Table III.

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TABLE III - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings Visual
302.	General Specification	3.2	MIL-F-13926

NOTE: The tests in Table III shall be conducted at a temperature of 73° ± 18°F.

4.5.2 Environmental. - Three head assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV, and shall then meet the requirements and tests in Table I.

TABLE IV - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperature (-80° to +160°F)	3.4.1	4.6.1
304.	Operating temperatures (-40° and +150°F)	3.4.2	4.6.2
305.	Operability, elevation input coupling	3.4.2, 3.5	4.6.2, 4.6.3

NOTE: The test for characteristic 305 shall be conducted at -40° and +150°F.

4.5.3 Verification, follow-up. - Three head assemblies shall be randomly selected as samples from production approximately at 25 percent; 50 percent and 75 percent completion interval of each production order or contract. Each sample shall meet the requirements and tests in Table V.

TABLE V - CLASSIFICATION OF DEFECTS

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Collimation, Pre-vibration daylight viewing area	3.7.1	4.6.4.1
307.	Collimation, pre-vibration passive viewing area	3.7.2	4.6.4.2
308.	Vibration "A"	3.9.1	4.6.7.1

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TABLE V - CLASSIFICATION OF DEFECTS - CONTINUED

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
309.	Collimation , post-vibration, daylight viewing area	3.10	4.6.8
310.	Collimation, post-vibration, passive viewing area	3.11	4.6.9
311.	Collimation, Pre-shcok daylight viewing area	3.7.1	4.6.4.1 ·
312.	Collimation, Pre-shock passive viewing area	3.7.2	4.6.4.2
313.	Shock	3.9.3	4.6.7.3
314.	Collimation, post-shock daylight viewing area	3.10	4.6.8
315.	Collimation, post-shock passive viewing area	3.11	4.6.9

NOTE: The tests in Table V shall be conducted at a temperature of 73° ± 18°F.

4.5.4 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted item has been approved (see 4.4.3).

4.5.5 Inspection Equipment. - Requirements for inspection equipment to perform examinations and tests are specified herein and in specific contract clauses as applicable. See Section 6 of this specification for general provisions applicable to inspection designs.

4.5.6 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.7 Contractor furnished inspection equipment.

4.5.7.1 Test equipment. - In conjunction with 4.5.6, the special inspection equipment in 4.5.7.1.1 and 4.5.7.1.2 shall be utilized in performing the applicable tests of 4.6.

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4.5.7.1.1 Sealing.-The pressure testing equipment to test sealing of the head assembly shall conform to the internal pressure test requirements of Drawing 8565556, Prwessure Tester-Two Station.

4.5.7.1.2 Performance. - Special test equipment to determine performance of the head assembly shall conform to the requirements of Drawing 11738140 (Final Acceptance Inspection Equipment) and all drawings pertaining thereto.

4.6 Methods of Inspection.

NOTE: At no time during the following tests shall the input coupling be allowed to strike hard against the travel stops.

4.6.1 Storage temperature. - This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope, this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-810 and the conditions of 3.4.1. The head assembly shall be placed in the test chamber and the temperature of the chamber reduced gradually (see NOTE below) to -80°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually raised (see NOTE below) to +160°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (73° ± 18°F) and the head assembly removed from the test chamber. The head assembly shall then be subjected to the visual, tactile, and audible examination of 4.6.7. The head assembly shall conform to the requirements of 3.4.1.

NOTE: The rate of temperature change in the test chamber shall not exceed 20 degrees per hour throughout the temperature cycling tests of 4.6.1 and 4.6.2.

4.6.2 Operating temperatures. - This test shall be performed on head assemblies produced as a repair part item. When the head assembly is provided with a complete periscope, this temperature test will not be required when the complete periscope is subjected to temperature cycling. The testing equipment utilized in this test shall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-810 and the conditions of 3.4.2. The head assembly shall be placed in the test chamber and the

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temperature of the chamber reduced gradually (see NOTE in 4.6.1) to -40°F. The head assembly shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F, the head assembly shall be subjected to the test in 4.6.3 to determine compliance with the applicable requirements of 3.5 for this temperature (-40°F). Upon completion of this test, the temperature of the test chamber shall be gradually raised (see NOTE in 4.6.1) to +150°F. The head assembly shall remain at this temperature for a period of 5 hours. At the completion of this 5 hour period, and while at +150°F, the head assembly shall again be subjected to the test in 4.6.3 to determine compliance with the applicable requirements of 3.5 for this temperature (+150°F). Upon completion of this test, the temperature of the test chamber shall be gradually reduced (see NOTE in 4.6.1) to room ambient temperature (73° ± 18°F) and the head assembly removed from the test chamber. Head assemblies which have passed the above tests shall be subjected to the test of 4.6.4.

4.6.3 Operability, elevation input coupling. - Standard torque measuring equipment capable of measuring the torque values specified in 3.5 shall be utilized for this test. An appropriate adapter to accommodate the coupling shall be used with the torque measuring equipment. The running torque required to rotate the coupling through its full range of travel in each direction, shall conform to the applicable requirements of 3.5 for the temperatures indicated.

4.6.4 Performance.-To determine compliance with 3.6, the following tests shall be performed with the head assembly positioned on the special test equipment specified in 4.5.7.1.2. Set-up of the test equipment and installation of the head assembly under test shall be performed in accordance with the set-up and calibration instructions contained on Drawing 8247391. The collimator Projector shall be focused at 1200 meters. Unless otherwise specified herein, these tests shall be performed at the daylight viewing area (Head Wedge 10541997).

4.6.4.1 Collimation, Pre-Vibration/Pre-Shock, daylight viewing area. - Prior to performing this test and the following tests of 4.6.4.2 and 4.6.4.3, disengage the index pin “D” from the index plate “E” of the test fixture and place the 10 second level on surface “C” of the leveling base. The bubble of the level shall be centered within its graduation lines. If it is not, then adjustment is required in accordance with the procedures specified in paragraphs 7 and 8 of the set-up instructions contained on Drawing 8247391. With the index pin “D” disengaged from the index plate and surface “C” of the leveling base level, view through Support Assembly Collimator 10548767 positioned at the daylight viewing area

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and observe the projected target image. The intersecting point of the horizontal and vertical reticle lines of the collimator shall b superimposed on the intersecting point of the horizontal and vertical target image lines within the tolerance specified in 3.7.1 as measured on the collimator's micrometer knobs. Record this measurement for use in the subsequent test of 4.6.8 and perform the following test of 4.6.4.2.

4.6.4.2 Collimation, Pre-Vibration/Pre-Shock, passive viewing area. - With the index pin "D" disengaged from the index plate "E't and surface "C" of the leveling base level, view through the auto-colli mater positioned at the passive viewing area (Head Wedge 10514254) and observe the projected target image The intersecting point of the horizontal and vertical reticle lines of the collimator shall be superimposed on the intersecting point of the horizontal and verticla target image lines within the the tolerance specified in 3.7.2 as measured on the collimator's micrometer knobs. Record this collimation measurement for use in the subsequent test of 4.6.9.

4.6.4.3 Collimation, unity power area. - Perform this test with the aid of the 26° Elevation Calibration Standard that conforms to Drawing 10548766. The calibration standard shall be set-up in accordance with the set-up instructions specified on Drawing 10547987 and then positioned at the unity power area (Head Window 10514256). With the index pi "D" disengaged from the index plate "E" and surface "C" of the leveling base level, view through the eyepiece of the 3X Telescope 5549108, contained in the calibration standard, and observe the boresight point (intersecting point of the horizontal and vertical reticle lines) of the telescope reticle with respect to the boresight point of the target image. The boresight point of the telescope reticle shall meet the collimation require- ments specified in 3.7.3, as measured with reference to the target image.

4.6.4.4 Backlash. - This test shall be performed subsequent to the test in 4.6.4.3. With the index pin "D" disengaged from the index plate "E", place the 10 second level on surface "C" of the leveling base. With backlash removed for direction of travel, rotate the leveling base by adjusting the leveling rod (Part #62 depicted on Drawing 8247391) until the bubble of the precision level is centered within its graduation lines without overtravel. View through the auto-collimator positioned at the daylight viewing area and position the horizontal and vertical reticle lines of the colli- mator coincident with the horizontal and vertical target image lines by means of the collimator's micrometer knobs. Continue rotation of the leveling base approximately 2° in the same

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direction of travel, then reverse rotation of the leveling base (backlash included) until the bubble of the precision level is re-centered within its graduation lines without overtravel. View through the collimator and observe for coincidence of the collimator's horizontal reticle line with respect to the horizontal target iamge line. Any adjustment of the collimator's micrometer knobs to re-obtain coincidence of the collimator's horizontal reticle line with respect to the horizontal target image line is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.7.7. Upon satisfactory completion of this test, disengage the leveling rod from the slot of the leveling base, engage the index pin "D" in the zero degree position on the index plate "E" and perform the following tests of 4.6.4.5 through 4.6.4.8 inclusive,

NOTE: Should overtravel occur during the test for backlash, the test shall be nullified and the procedure specified for backlash shall again be performed.

4.6.4.5 Excursion range.-With the input coupling positioned as specified in 3.6b and the index pin "D" engaged in the zero degree position on the index plate "E", view through the auto-collimator positioned at the daylight viewing area and, if necessary, position the horizontal and vertical reticle lines of the collimator coincident with the horizontal and vertical target image lines by means of the collimator's micrometer knobs. When coincidence has been established, view through the collimator to the reflector "H" on the test fixture. Observe the collimator reticle and note the position of the boresight point (intersecting point of the horizontal and vertical reticle lines) with respect to its reflected image. Disengage the index pin from the zero degree elevation position on the index plate. While observing the boresight point of the real and reflected image, rotate the input coupling and engage the index pin, first in the 22 degree elevation position and then in the 18 degree depression position on the index plate. While the index pin is engaged at each position (22°EL and 18°DP), the noted position of the boresight point with respect to its reflected image shall not deviate in elevation or depression respectively to determti.ne conformance with 3.7.4 (see 4.6.4.7 and 4.6.4.8 when performing this test).

4.6.4.6 Coupling travel. - Rotate the coupling to its maximum elevation, then to its maximum depression stop position. While the coupling is at each stop, back off the coupling for at least 1/4 rotation and then again return the coupling to the stop. Perform this a minimum of three times. During these rotations, the coupling

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shall show no evidence of irregularity of movement (sponginess or friction) while approaching or reaching the stop. Contact with the stop shall be solid such that a distinct metal click can be heard to verify that coupling rotation is being limited only by means of contact with the coupling stop pin and not by contact of other components (optical or mechanical) to be in compliance with 3.7.4.1.

4.6.4.7 Plumb travel. -This test may be performed at the same time that the test of 4.6.4.5 is performed. While observing the boresight point (intersecting point of the horizontal and vertical reticle lines) of the real and reflected image, rotate the input coupling so that the line of sight is deflected from 22° above to 18° below the noted position of the boresight point established in 4.6.4.5. The boresight point of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.5 for the positions indicated as measured on the collimator reticle (see 4.6.4.8 when performing this test).

4.6.4.8 Elevation accuracy. - Perform this test in conjunction with test in (plumb travel). The horizontal reticle lines of the real and reflected image shall not deviate beyond the applicable value specified in 3.7.6 for the positions indicated.

4.6.4.9 Image tilt. - This test shall be performed with the index pin “D” engaged in the zero degree position on the index plate "E". view through the auto-collimator positioned at the Head Wedge 10541997 and position one extremity of the collimator’s vertical reticle line coincident to the vertical target image line. The image of the vertical target line shall be parallel to the collimator’s vertical reticle line within the tolerance specified in 3.7.8, as measured on the target. Repeat the above procedure with the auto-collimator positioned at the Head Wedge 10514254. The image of the vertical target line shall be parallel to the collimator’s vertical reticle line within the tolerance specified in 3.7.8, as measured on the target.

4.6.5 Resolution, datlight and passive viewing areas. - The test equipment specified in 4.6.4 is not required for this test (4.6.5) and the test in 4.6.6. Position the head assembly on its side with side “L” of the input coupling positioned at 22° 30 minutes with respect to the three mounting pads. This test shall be conducted utilizing an observation telescope of at least 20 power with an entrance pupil as specified in 3.8.1, a resolving power chart representing the angular subtense for the seconds of arc specified and a supporting medium capable of receiving the observation telescope as well as the head assembly to be tested. Construction

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and positioning of the resolving power chart shall comply with the "Resolution test" specified in MIL-I-13830 and shall contain four line sets as shown in Figure I therein. Position the observation telescope at the Head Wedge 10541997. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3,8.1. Repeat the above procedure with the observation telescope positioned at the Head Wedge 10514254. The focus of the telescope shall be established at one best focus setting for viewing of the detectable line structure and proper line count in all four meridians of the resolving power chart. Resolution shall meet the requirement specified in 3.8.1

4.6.6 Resolution, unity power system. - Perform this test subsequent to 4.6.5, utilizing an observation telescope of at least 3 power with an entrance pupil as specified in 3.8.2 and a resolving power chart representing the angular substense for the seconds of arc specified. Construction and positioning of the resolving power chart shall be as specified in 4.6.5. Resolution shall be within the tolerance specified in 3.8.2, as determined by viewing through the observation telescope when positioned at the Head Window 10514256.

4.6.7 Vibration/Shock.

4.6.7.1 Vibration "A". - This test shall be performed on head assemblies procured as a repair part item. when the head assembly is provided with a complete periscope, this vibratio test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 and 4.6.4.2 to determine compliance with 3.7.1 and 3.7.2. Subsequent to the test of 4.6.4.1 and 4.6.4.2, the head assembly shall be vibrated in accordance with MIL-STD-81O, Method 514.2, Procedure VIII, except the test level and frequency range shall be as specified in 3.9.1. Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment testshall be in accordance with the "Test Conditions" specified in the "General Requirements" of MIL-STD-81O. The vibrating equipment shall also be capable of vibrating the head assembly along three mutually perpendicular axes while the head assembly is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing 11747439). The head assembly shall be clamped on the vibrating equipment by means

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of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The head assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period specified in 3.9.1. During this test, the input coupling shall be immobilized to any position that shall prevent it from contactingthe travel stops. Immediately following the vibration cycle, and prior to the examinatio for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 inclusive and 4.6.10 through 4.6.12 inclusive.

4.6.7.2 Vibration “B”. - This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope, this vibration test will not be required when the complete periscope is to be vibrated. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 and 4.6.4.2 to determine compliance with 3.7.1 and 3.7.2. Subsequent to the test of 4.6.4.1 and 4.6.4.2, the head assembly shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure VIII, except the test level and frequency range shall be as specified in 3.9.2. Vibration shall be applied along each of three mutually perpendicular axes of the head assembly. Vibrating equipment capable of providing the test level and frequency ranges specified above shall be utilized for this test. The vibrating equipment shall be in accordance with the “Test Conditions” specified i nthe “General Requirements” of MIL-STD-810. The vibrating equipment shall also be capable of vibrating the head assembly along three mutually perpendicular axes while the head assembly is maintained in it's normal mounting or operating position (as depicted in the front view on Sheet 1 of Drawing 11747439). The head assembly shall be clamped on the vibrating equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The head assembly shall then be vibrated in accordance with the test procedure specified above and swept logarithmically for the time period

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specified in 3.9.2. During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Immediately following the vibration cycle, and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.5.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9.2, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 inclusive and 4.6.10 through 4.6.12 inclusive.

4.6.7.3 Shock. - This test shall be performed on head assemblies procured as a repair part item. When the head assembly is provided with a complete periscope, this shock test will not be required when the complete periscope is subjected to shock. Prior to performing this test, the head assembly shall be subjected to the test procedure specified in 4.6.4.1 to determine compliance with 3.7.1 and 3.7.2. The shock equipment and test procedure shall be in accordance with MIL-STD-810, Method 516.2, Procedure I, except conditions such as “G” values, time duration and wave form shock pulse shall be as specified in 3.9.3. The head assembly shall be clamped on the shock equipment by means of an appropriate adapter having the same interface features as Mount Periscope: M118E1. The head assembly shall then be shocked in accordance with the test procedure specified above, except the application of shock along the vertical axis shall be in one direction only (normal operating position of the head assembly). During this test, the input coupling shall be immobilized to any position that shall prevent it from contacting the travel stops. Immediately following the shock test and prior to the examination for physical defects resulting from poor workmanship and materials, the head assembly shall be subjected to the tests in 4.6.8 and 4.6.9. Caution shall be exercised during transfer of the head assembly. Subsequent to the tests in 4.6.8 and 4.6.9, the head assembly shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine compliance with 3.9.3, prior to being subjected to the remaining tests specified herein. A visual

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examination shall be made for missing or damaged parts. A tactile examination shall be made for functioning of moving parts, for irregularities of movement, friction or looseness. An audible examination shall also be made of the head assembly for loose internal optical and mechanical components. Any one of the possible defects defined above shall be cause to consider a head assembly defective. Head assemblies which have passed the above examinations shall be subjected to the remaining tests specified in 4.6.3, 4.6.4.4 through 4.6.6 inclusive and 4.6.10 through 4.6.12 inclusive.

4.6.8 Collimation, Post-Vibration/Post-Shock, daylight viewing area. - Perform this test utilizing the procedure in 4.6.4.1. Record the collimation measurement. Any difference between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.4.1 shall not exceed the tolerance specified in 3.10

4.6.9 Collimation, Post-Vibration/Post-Shock passive viewi~ area. - Perform this test utilizing the procedure in 4.6.4.2. Record the collimation measurement. Any differncece between the recorded measurement obtained in this test and the recorded measurement obtained in 4.6.4.2 shall not exceed the tolerance specified in 3.11

4.6.10 Sealing. - This portion of the test shall be performed utilizing a pressure gage capable of reding between 0.25 and 0.50 pound per square inch gage (psig) and a dew point tester. The following test sequence shall be adhered to. Check the internal pressure of the head assembly utilizing the pressure gage specified herein. Then, test the dew point of the interior atmosphere of the head assembly by means of the dew point testers. Interior atmosphere of the head assembly shall conform to the requirements of 3.12. Upon satisfactory completion of this portion of the test, internal pressurizing of the head assembly shall be performed with special testing equipment conforming to 4.5.2.3.1. Pressurize the head assembly with dry nitrogen gas, having a dew point at least as low as -25°F, to 5 ± 0.10 psig in accordance with the procedure for internal pressure testing on Drawing 8565556. Retain the applied pressure in the head assembly for a minimum of 2 hours. During the first hour, the input coupling shall be rotated throught one full excursion of its articulation range. At the end of the 2 hour period there shall be no evidence of leadage in excess of 0.2 psi as indicated on the pressure gage of the testign equipment, to deter- mine compliance with 3.12. Upon satisfactory completion of the sealing test, the head assembly shall then be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the head assembly shall be periodically sampled through the dew point tester

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until the atmosphere shows a dew point at least as low as -25°F. When this atmosphere is attained within the head assembly, the head assembly shall then be sealed at the pressure specified in 3.12,

4.6.11 Cleanliness. - The optical surfaces of the head assembly shall be inspected for cleanliness with the unaided eye. While viewing through each end of the module, inspection for defects shall be made by the technique of shadowing. while viewing through each Head Wedge (10541997 and 10514254), inspection for dirt shall be made by viewing an illuminated field provided by 2 each 15 watt desk flourescent lamps held close to the Head Window 10514255. The head assembly shall conform to the requirements of 3.13a and 3.13b.

4.6.12 Interchangeability.

4.6.12.1 Head Assembly with modules (Elbow Assembly, Passive and Body Assembly, Daylight). -The cavity at the Head Wedge 10541997 shall be inspected utilizing the maximum and minimum interchangeability gages conforming to Drawings 11732107 and 11732108 respectively, to determine compliance with 3.2a. Upon successful completion of this portion of the test, the cabity at the Head Wedge 10514254 shall be inspected utilizing the minimum and maximum interchangeability gages conforming to Drawings 11729723 and 11732101 respectively, to determine overall compliance with 3.2a.

4.6.12.2 Head Assembly with mount. - The interchangeability test of the head assembly with its mount shall be performed utilizing the maximum and minimum interchangeability gages conforming to Drawings 11732155 and 11732156. The head assembly shall conform to the requirements of 3.2b.

5 PACKAGING

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11747439. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The Head Assembly 11747439 is a modular unit of the Tank Gunner's Periscope M32CE1 primarily used in the fire control system of the 165mm Gun, Combat Engineer Vehicle: M728.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules

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c. Applicable packaging data sheet number (see 5.1).

d. Selection of applicable levels of preservation, packaging and packing.

e. Applicable stock number.

f. Requirements for First Article Submission, including data requirements for certificates of conformance for materials.

6.3 Inspection equipment design.

6.3.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications of SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the Technical Data Package (TDP).

6.3.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with tile “Methods of Inspection” specified in 4.6 of this specification and with the “Test Conditions” specified in the “General Requirements” of MIL-STD-810. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction materials and dimensions shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when non-conforming. Construct-ion shall be such as to facilitate routine calibration of test equipment.

6.3.3 Submission of designs for approval. - Contractor designs shall be approved by the Government prior to fabricating or pro-curing the- equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAF-I, Dover, NJ 07801.

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When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number
b. The contract item (name, model number, etc.)
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Drawings. - Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picacinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

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MIL-H-48620 HEAD ASSEMBLY· 11747439

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